

# Used-car appraisal

## Preliminary insight development

REVISION DRAFT · All numerical results require original-XLSM verification in Microsoft Excel.

| PREVIOUS CSV ROWS | CSV SCREENED ROWS | DRAFT MAIN TEXT |
|-------------------|-------------------|-----------------|
| 1,005,000         | 998,849           | 1,000 words     |

### 1. Introduction

The client operates an Indian online used-car marketplace where faster, defensible appraisals could improve cash offers and listing-price checks [1]. Overvaluation threatens purchasing margins; undervaluation risks losing sellers. This milestone examines vehicle age, usage and brand segment to prioritise subsequent analysis. It provides preliminary evidence and a development plan rather than a completed pricing model. Only the fixed randomised dataset is analysed, using the descriptive, probability and sampling concepts introduced in Weeks 1–3 [2–4]. Tables 1–4 contain preliminary CSV summaries; Figures 1–4 await actual Excel chart generation.

### 2. Data Summaries and Descriptive Statistics

#### 2.1 Data definition and preparation

The dataset [6] contains 1,005,000 listings and 20 variables. Car\_Price, recorded in Indian rupees, is the outcome; Registration\_Age and Kms\_Driven measure supplied age in years and usage in kilometres. Table 2 distinguishes nominal labels, ownership categories, numerical measurements and binary flags. In every observation, Year plus Registration\_Age equals 2025. The supplied age is therefore retained as a snapshot measure, rather than recalculated using today's date; including both age and Year later would duplicate information.

Trimming Brand corrects 10,040 labels and recovers 2,334 luxury records that an uncleaned lookup would misclassify. Luxury comprises Audi, BMW and Mercedes; the other ten identified brands constitute non-luxury. Explicit spelling, case and whitespace corrections standardise 20,123 Fuel\_Type entries. Unknown categories remain visible rather than being guessed. This export contains neither blank cells nor exact duplicates, including after label cleaning; those checks do not establish that the original source was complete [5]. Forty zero readings across three mechanical variables are flagged for verification, without imputing unused predictors.

Fuel type and transmission are Unknown for 7.83% and 8.00% of full-file records respectively; dropping those rows would discard usable price, age and usage observations. These unresolved categories will be reported separately when studying additional predictors.

The primary cohort provisionally screens 6,151 listings exceeding 1,000,000 kilometres, leaving 998,849 observations. This threshold follows the guide's example of implausible usage but is an appraisal-screening decision, not proof of error. All records remain in the full-sample comparison. High prices are retained: an IQR flag alone cannot establish an incorrect valuation. Table 1 records decisions and denominators.

#### 2.2 Distribution and typical values

The primary mean price is INR 1,015,954, compared with a median of INR 673,944. The substantial gap and previously observed right tail suggest positive skew. The price IQR is INR 877,727 and sample SD is INR 1,141,023; dispersion is commercially material. The median is preferable for describing a typical listing because extreme prices pull the mean upward [2]. However, it is not a vehicle-specific appraisal, and the mean remains informative for portfolio averages.

Average age is 13.00 years and median usage is 152,632 kilometres. Table 3 includes means, medians, modes, sample SDs, quartiles and ranges for the full file and screened groups. At original XLSM precision, price and usage have no mode: all values are unique.

# Analysis, conclusions and next steps

## 2.3 Relationships with age and usage

Figures 1–2 must be rebuilt from 5,000 screened listings selected in Excel using RAND(), fixed random values, sorting and retained original row numbers. They are absent pending verification. Correlations must use unrounded prices and every eligible pair. The earlier CSV age-price correlation was  $-0.0012$  overall and negligible within both segments. Age-group means must confirm whether the original XLSM supports that provisional pattern.

Usage has a weak negative linear association with price:  $r = -0.0509$  overall,  $-0.0521$  for luxury and  $-0.0505$  for non-luxury. Previous CSV summaries show declining mean prices across usage deciles, alongside substantial overlap in the earlier plots. Hence usage may inform further investigation but cannot support precise offers alone. Restoring high-usage records changes the overall correlation to  $-0.0499$ ; the interpretation survives this sensitivity check. Pearson correlation describes linear association, not causation or all possible nonlinear structure [2].

An additional diagnostic removes prices above the upper IQR fence of INR 2,560,035. Usage correlation remains weak at  $-0.0516$ , so the principal conclusion is not driven solely by unusually expensive listings; this diagnostic is not the adopted sample.

## 2.4 Brand-segment comparison

The screened cohort contains 230,761 luxury and 768,088 non-luxury listings. Their mean prices are INR 1,014,112 and INR 1,016,508 respectively: a luxury difference of  $-0.236\%$ . Medians differ by only INR  $-435$ , or  $-0.065\%$ . The earlier CSV distributions overlap substantially, and both groups have comparable ages and usage. These summaries provide no commercially substantial unconditional luxury premium in this version. They neither establish equality in the population nor rule out differences conditional on other characteristics. Empirical segment shares describe these listings, not India's market composition [3].

## 3. Conclusion and Development Plan

The useful preliminary findings are strong price skew, considerable dispersion, negligible age association and a weak usage association. The evidence supports cautious investigation, not automatic depreciation deductions or a fixed luxury surcharge. Listing prices are not verified transaction prices; city coverage, Unknown labels and unresolved measurements limit generalisation. Randomisation may weaken real-world relationships. A million observations can improve precision without removing selection bias: the Central Limit Theorem concerns sample means, not a normal distribution of individual prices or representative sampling [4].

For Milestone 2, define  $\Delta = \mu_{\text{luxury}} - \mu_{\text{non-luxury}}$  and examine  $H_0: \Delta = 0$  against the two-sided alternative  $H_1: \Delta \neq 0$ . Report the estimated mean difference in INR, its percentage relative to non-luxury prices, and a confidence interval using subsequently taught methods. Interpret uncertainty and practical relevance together.

Investigate age and usage using within-segment summaries, age groups and usage deciles, checking curvature and whether aggregate patterns persist within brands. State denominators and revisit the mileage screen. For the final report, assess the other supplied variables, including model, fuel type, transmission, engine characteristics, ownership history, city and service history. Compare a pooled model with brand-specific models using the same reserved observations. Report mean absolute error and root mean squared error in INR, overall, by brand and for expensive listings; compare against a simple benchmark and avoid information leakage. Verify flagged measurements and transaction-price limitations before operational recommendations. M1 proposes these analyses without performing hypothesis tests or fitting predictive models. Uncertain appraisals should remain subject to human review.

# Evidence A · Preparation and audit

**Table 1a. Provisional CSV audit; rerun against original XLSM**

Counts on this page come from the earlier Python/CSV analysis. None has been independently verified in Excel against the original XLSM. Retain the raw worksheet and Dataset!B6, rerun each check, and replace these counts before submission.

| Audit item                                 | Observed count | Treatment / rationale                                                             |
|--------------------------------------------|----------------|-----------------------------------------------------------------------------------|
| Source rows                                | 1,005,000      | Complete supplied CSV; source hash verified.                                      |
| Blank cells                                | 0              | 0 in this export; not proof the original source was always complete.              |
| Exact duplicate rows                       | 0              | Compared all 20 supplied fields; keep first rule would apply only if found.       |
| Duplicates after label cleaning            | 0              | Rechecked all supplied fields after canonicalisation.                             |
| Brand labels corrected                     | 10,040         | Trim leading/trailing spaces; 26 labels become 13.                                |
| Luxury records recovered by Brand trimming | 2,334          | 2,334 records otherwise misclassified by an exact uncleaned lookup.               |
| Fuel labels corrected                      | 20,123         | Case/space correction and explicit hybridd/electrik mappings; 12 labels become 6. |
| Kms_Driven > 1,000,000                     | 6,151 (0.612%) | Flag and screen provisionally; retained in full-sample sensitivity analysis.      |
| Primary cohort rows                        | 998,849        | No other record exclusions; all key variables valid in this cohort.               |
| Zero Mileage_kmpl / Engine_CC / Horsepower | 5 / 12 / 23    | Flag unresolved auxiliary readings; do not call them blanks or impute them.       |
| High price IQR flags in primary cohort     | 70,457         | Diagnostic only; no price-based exclusion in primary results.                     |

**Table 1b. Unknown categories retained in the full file**

| Variable     | Unknown entries | Share of 1,005,000 |
|--------------|-----------------|--------------------|
| Fuel_Type    | 78,733          | 7.83%              |
| Transmission | 80,385          | 8.00%              |
| Color        | 80,419          | 8.00%              |
| City         | 80,430          | 8.00%              |

**Table 1c. Cohort reconciliation after Brand normalisation**

| Stage              | All       | Luxury  | Non-luxury |
|--------------------|-----------|---------|------------|
| Full supplied file | 1,005,000 | 232,172 | 772,828    |
| High-usage screen  | 6,151     | 1,411   | 4,740      |
| Primary cohort     | 998,849   | 230,761 | 768,088    |

# Evidence B · Variables and units

Table 2. Definitions, measurement types and analytical roles

| Variable         | Type / unit                      | Meaning and treatment                                                 |
|------------------|----------------------------------|-----------------------------------------------------------------------|
| Brand            | Nominal                          | Vehicle manufacturer; cleaned before segment construction.            |
| Model            | Nominal                          | Model name; 39 observed labels.                                       |
| Year             | Integer / year                   | Manufacturing year, 2000–2024. Age + Year = 2025 in every row.        |
| Mileage_kmpl     | Continuous / km per litre        | Fuel efficiency; five zero values flagged.                            |
| Engine_CC        | Continuous / cm³                 | Engine displacement; twelve zeros require contextual checking.        |
| Horsepower       | Continuous / reported horsepower | Engine power output; twenty-three zero readings flagged.              |
| Fuel_Type        | Nominal                          | Petrol, Diesel, CNG, Electric, Hybrid, Unknown after correction.      |
| Transmission     | Nominal                          | Manual, Automatic, Unknown.                                           |
| Owner_Type       | Ordinal category                 | First, Second, Third, Fourth+ owner category.                         |
| Color            | Nominal                          | Vehicle colour; Unknown preserved as an unresolved label.             |
| City             | Nominal                          | Registration city; eight named cities and Unknown.                    |
| Kms_Driven       | Continuous / km                  | Total distance travelled; key usage variable.                         |
| Insurance_Valid  | Binary / 0 or 1                  | Recorded insurance-status flag; 1 indicates valid insurance.          |
| Service_History  | Binary / 0 or 1                  | Recorded service-history availability flag.                           |
| Accidents        | Integer / count                  | Number of reported accidents, 0–5.                                    |
| Tax_Paid         | Binary / 0 or 1                  | Recorded road-tax-payment flag.                                       |
| Number_of_Doors  | Integer / count                  | Door count; observed values 2, 4, 5.                                  |
| Seats            | Integer / count                  | Seating capacity; observed values 2, 4, 5, 7.                         |
| Registration_Age | Integer / supplied years         | Key age measure, 1–25; retain the supplied snapshot definition.       |
| Car_Price        | Continuous / INR                 | Reported price in rupees; outcome, not verified transaction proceeds. |

Source: course dataset description shown in [5, PDF p. 8] and the user’s Dataset worksheet screenshot [6]. INR denotes Indian rupees. Age uses the supplied year units and snapshot; the Year + Registration\_Age = 2025 identity was checked only in the previous CSV and requires original-XLSM verification. Auxiliary zero measurements are flagged but retained; the original-XLSM workbook must record their formula-based summaries before submission.

# Evidence C · Numerical summaries

Table 3. Full file and primary cohort, overall and by segment

F = full supplied file (labels standardised). S = primary cohort (Kms\_Driven ≤ 1,000,000). Price and usage are scaled by 1,000; sample SD uses n – 1. These numerical columns are previous CSV calculations, not Excel-verified original-XLSM results; rounding is for display only.

Table 3a. Car\_Price (INR thousands)

| Cohort        | Valid n   | Mean    | Median | SD      | Min  | Max      | Mode†    |
|---------------|-----------|---------|--------|---------|------|----------|----------|
| F: All        | 1,005,000 | 1,014.8 | 673.1  | 1,140.1 | 10.4 | 46,802.5 | No mode† |
| F: Luxury     | 232,172   | 1,013.0 | 672.8  | 1,129.1 | 11.5 | 46,802.5 | No mode† |
| F: Non-luxury | 772,828   | 1,015.3 | 673.2  | 1,143.4 | 10.4 | 38,036.4 | No mode† |
| S: All        | 998,849   | 1,016.0 | 673.9  | 1,141.0 | 10.4 | 46,802.5 | No mode† |
| S: Luxury     | 230,761   | 1,014.1 | 673.6  | 1,130.3 | 11.5 | 46,802.5 | No mode† |
| S: Non-luxury | 768,088   | 1,016.5 | 674.0  | 1,144.2 | 10.4 | 38,036.4 | No mode† |

Table 3b. Registration\_Age (years)

| Cohort        | Valid n   | Mean  | Median | SD   | Min  | Max   | Mode† |
|---------------|-----------|-------|--------|------|------|-------|-------|
| F: All        | 1,005,000 | 13.00 | 13.00  | 7.21 | 1.00 | 25.00 | 6     |
| F: Luxury     | 232,172   | 13.02 | 13.00  | 7.21 | 1.00 | 25.00 | 18    |
| F: Non-luxury | 772,828   | 13.00 | 13.00  | 7.21 | 1.00 | 25.00 | 19    |
| S: All        | 998,849   | 13.00 | 13.00  | 7.21 | 1.00 | 25.00 | 6     |
| S: Luxury     | 230,761   | 13.02 | 13.00  | 7.21 | 1.00 | 25.00 | 18    |
| S: Non-luxury | 768,088   | 13.00 | 13.00  | 7.21 | 1.00 | 25.00 | 19    |

Table 3c. Kms\_Driven (thousand km)

| Cohort        | Valid n   | Mean  | Median | SD    | Min | Max     | Mode†    |
|---------------|-----------|-------|--------|-------|-----|---------|----------|
| F: All        | 1,005,000 | 203.0 | 153.5  | 175.9 | 4.5 | 6,259.5 | No mode† |
| F: Luxury     | 232,172   | 202.9 | 153.2  | 175.7 | 4.5 | 4,129.8 | No mode† |
| F: Non-luxury | 772,828   | 203.1 | 153.6  | 175.9 | 4.7 | 6,259.5 | No mode† |
| S: All        | 998,849   | 196.2 | 152.6  | 150.7 | 4.5 | 999.9   | No mode† |
| S: Luxury     | 230,761   | 196.1 | 152.4  | 150.9 | 4.5 | 999.5   | No mode† |
| S: Non-luxury | 768,088   | 196.2 | 152.7  | 150.6 | 4.7 | 999.9   | No mode† |

† Car\_Price and Kms\_Driven: No mode — all values unique, according to the user’s original-XLSM check; independent Excel confirmation is pending. Age modes shown here are provisional CSV values. Do not use CSV precision to establish modes in the XLSM.

Table 3d. Primary-cohort quartiles and interquartile ranges

| Variable / unit       | All: Q1 / Q3 / IQR      | Luxury: Q1 / Q3 / IQR   | Non-luxury: Q1 / Q3 / IQR |
|-----------------------|-------------------------|-------------------------|---------------------------|
| Price / INR thousands | 365.7 / 1,243.4 / 877.7 | 365.0 / 1,246.6 / 881.5 | 365.9 / 1,242.6 / 876.6   |
| Age / years           | 7.0 / 19.0 / 12.0       | 7.0 / 19.0 / 12.0       | 7.0 / 19.0 / 12.0         |
| Usage / thousand km   | 92.4 / 251.7 / 159.3    | 92.4 / 251.4 / 158.9    | 92.4 / 251.8 / 159.4      |

Quartiles use linear interpolation, equivalent to Excel QUARTILE.INC / PERCENTILE.INC. The earlier evidence workbook contains CSV-derived results, not original-XLSM formula evidence. The new verification template is an unexecuted checklist; it is not a completed analysis workbook.

# Evidence D · Relationships

**Table 4. Provisional CSV correlations: Excel verification pending**

| Cohort / segment    | Eligible pairs | r: age-price | r: usage-price |
|---------------------|----------------|--------------|----------------|
| F: All              | 1,005,000      | -0.0012      | -0.0499        |
| F: Luxury           | 232,172        | -0.0009      | -0.0508        |
| F: Non-luxury       | 772,828        | -0.0013      | -0.0497        |
| S: All              | 998,849        | -0.0012      | -0.0509        |
| S: Luxury           | 230,761        | -0.0008      | -0.0521        |
| S: Non-luxury       | 768,088        | -0.0013      | -0.0505        |
| Extra price screen* | 928,392        | -0.0005      | -0.0516        |

\* Diagnostic only: additionally omit 70,457 prices above INR 2,560,035 ( $Q3 + 1.5 \times IQR$ ). Primary estimates retain those prices. No hypothesis tests are performed.

## Figures 1-2 · Excel scatterplots pending

The previous Python scatterplots have been withdrawn from this revision draft. Removing their pandas caption would not make them Excel-generated. No replacement scatterplot has yet been sampled or drawn in Microsoft Excel.

### Required procedure

Preserve an original worksheet-row identifier. For every eligible row, generate RAND() in Excel, immediately paste the random column as values, and save. Sort the entire eligible table by that fixed random column, using the original row identifier to break ties. Retain the first 5,000 rows and their original identifiers.

### Required chart evidence

In Excel, create XY scatterplots of Registration\_Age versus Car\_Price and Kms\_Driven versus Car\_Price from the same fixed sample. Record eligible and sampled segment counts, workbook version, selection date and chart data ranges. If using a logarithmic price axis, disclose it; calculate CORREL on untransformed full-cohort values.

### Caption to complete after Excel execution

Figures 1-2. Excel RAND() sample; frozen values; eligible n = [verified]; sampled n = 5,000; luxury n = [verified]; non-luxury n = [verified]. Original worksheet rows and both chart data ranges are retained in Sample\_5000. The identical fixed sample is used in both charts.

# Evidence E · Excel charts to complete

Figures 3–4 have also been withdrawn from this draft. They must be recreated in Excel from the verified original-XLSM cohort; the previous Python images are not retained as report evidence.

| Figure    | Planned Excel calculation and display scope                                                                                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Figure 3  | Create equal-width INR 250,000 price bins in Excel. Use each complete segment as its denominator. If the plot stops at INR 5 million, disclose verified counts and shares beyond that limit; retain those prices in the summary statistics.                   |
| Figure 4a | Use an Excel PivotTable to report count and average original-precision price at each supplied age and within each segment. Plot the verified group means; retain group sizes and note that connecting lines are not fitted models.                            |
| Figure 4b | Calculate pooled usage-decile cutoffs with PERCENTILE.INC, retain the cutoffs, and allocate boundary ties consistently. In a PivotTable, summarise mean price, mean usage and n within each segment/decile. Group means do not describe individual variation. |

## Report sufficiency and workbook evidence

The final report must display the verified summary tables, scatterplots, grouped patterns, sample sizes and treatment decisions. Readers should understand its key findings without opening a companion workbook. The evidence workbook supports verification rather than substituting for missing report evidence.

Preserve the original data and Dataset!B6; retain cleaning formulas and a rule-by-rule audit log; show formulas for full, luxury and non-luxury groups; save fixed sample row numbers and actual Excel chart ranges. No verified original-XLSM evidence workbook has yet been produced.

# References and source verification

[1] UNSW Business School (2026). COMM5000 T3 2026 Assessment Guide Milestone 1 – Used Cars. PDF pp. 6, 8–11: business scenario, M1 tasks, structure, submission requirements and marking rubric.

[2] UNSW Business School (2026). COMM5000–Week 1–T326. PDF pp. 27–55: data quality, sampling, graphical and numerical summaries, dispersion and Pearson correlation.

[3] UNSW Business School (n.d.). COMM5000 W2 Lecture. PDF pp. 10, 17–22, 32–34: empirical and conditional probability; probability distributions.

[4] UNSW Business School (n.d.). COMM5000 W3 Lecture. PDF pp. 4–11: descriptive/inferential statistics, estimators, sampling distributions and the Central Limit Theorem. Historical administrative dates are not used.

[5] UNSW Business School (2026). COMM5000\_Course-Dataset-M1-intro-T326. PDF pp. 2–8: project scope, workbook use, dataset-description screenshot and randomisation guidance.

[6] COMM5000 course dataset (2026). COMM5000-Used\_Car\_Price\_Prediction.xlsm, the intended source for the revised analysis. The original file is not currently readable in this workspace. The supplied Dataset screenshot records B6 randomisation on 2026-10-07 at 06:20:30, timezone unspecified. The numerical values in this draft are earlier CSV-derived results and are not original-XLSM Excel verification.

Course-material access: <https://moodle.telt.unsw.edu.au/> (login required). Exact file names and physical PDF page locations are given above. Individual Moodle resource URLs were not provided. The original publisher address is recorded in the supplied dataset description; no external observations or market reports were used.

## Source and completion record

| Item                        | Status                                                                                                                               |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Intended numerical source   | The same original randomised XLSM, using stored numeric precision. Do not substitute a CSV or regenerate randomisation.              |
| Randomisation record        | Preserve Dataset!B6 exactly; screenshot timestamp 2026-10-07 06:20:30.                                                               |
| Price / usage modes         | No mode — all values unique: user-reported original-XLSM check. Independent Excel confirmation pending.                              |
| Other numbers in this draft | Previous Python calculations on a CSV. All counts, descriptive statistics and correlations require original-XLSM Excel verification. |
| Graphs                      | Previous Python charts withdrawn; actual Excel sampling and replacement charts are still pending.                                    |
| Evidence workbook           | Verification template is unexecuted; original data, retained formulas, frozen sample and Excel charts must be supplied.              |
| Main-report word count      | 1,000 words including section headings; title, metadata, tables, figures/captions and references excluded.                           |